AtomS3R-M12

SKU:C126-M12

















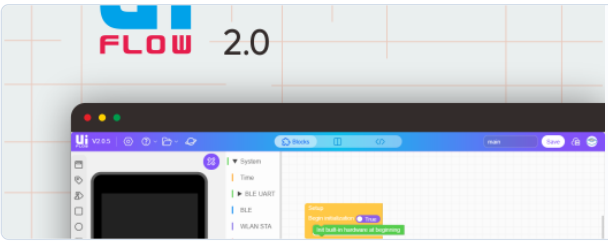




Description

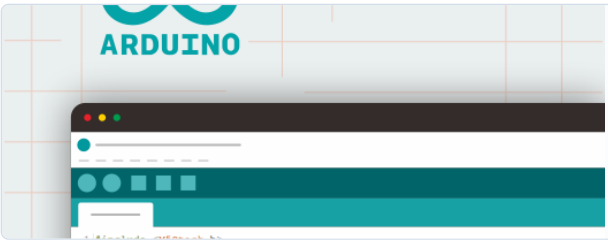
AtomS3R-M12 is an IoT programmable controller integrated with an **M12 camera**, equipped with a 3-megapixel **OV3660 wide-angle** camera, supporting high-resolution image capture. It uses the **ESP32-S3-PICO-1-N8R8** as the main controller, integrating Wi-Fi functionality, 8MB on-chip **Flash**, and 8MB **PSRAM**. It features a 5V to 3.3V power management circuit and includes a three-axis **BMM150** geomagnetic sensor and a six-axis **BMI270** attitude sensor. The product comes with **UVC** (USB Video Class) firmware, allowing it to be used as a driver-free USB camera, enabling users to use the camera directly via USB without installing drivers, and it has infrared emission control functionality. Compared to previous ATOM series host products, the **3D antenna** of **AtomS3R-M12** is enhanced, providing better performance and higher stability. The product also includes a **USB Type-C** interface for power supply and firmware download, and features a **HY2.0-4P** expansion port. The bottom is designed with six **GPIO** and power pins for easy expansion. With dimensions of only **24.0 x 24.0 x 22.1mm**, it is suitable for embedded scenarios such as IoT device monitoring and educational development tools.

Tutorial



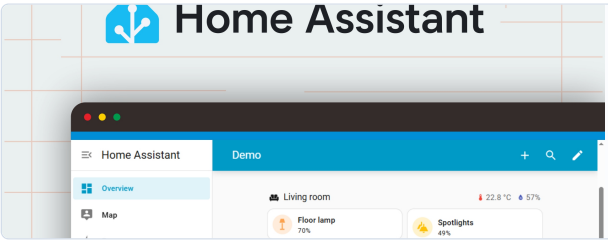
UiFlow2

This tutorial shows you how to control the AtomS3R-M12 using the UiFlow2 graphical programming platform.



Arduino IDE

This tutorial will introduce you to programming and controlling the AtomS3R-M12 device using the Arduino IDE



Home Assistant

This tutorial explains how to integrate AtomS3R-M12 into Home Assistant.

Features

- Integrated ESP32-S3-PICO-1-N8R8 main controller
- 3-megapixel OV3660 camera
- Nine-axis sensor system (BMI270 six-axis + BMM150 three-axis geomagnetic sensor)

- 8MB Flash and 8MB PSRAM
- Supports infrared emission control functionality
- Expandable pins and interfaces
- Development Platform
  - Arduino IDE
  - ESP-IDF
  - PlatformIO

Learn

BMM150 Magnetic Field Interference

Products with magnets may interfere with the BMM150 magnetic field sensor, causing abnormal readings. When used with an M5 master control device containing a magnet, the magnet needs to be removed, and at the same time, the BMM150 sensor should be kept away from strong magnetic fields.

Includes

- 1 x AtomS3R-M12

Applications

- Driver-free USB camera
- Motion detection and orientation awareness
- Smart device control
- IoT applications

Specifications

Specification	Parameter
SoC	ESP32-S3-PICO-1-N8R8 @ Xtensa® 32-bit LX7 dual-core processor, clock frequency 240MHz
USB	USB OTG, USB Serial/JTAG
Flash	8MB
PSRAM	8MB
Wi-Fi	2.4 GHz Wi-Fi
Six-axis attitude sensor (BMI270)	Accuracy: 0.05% (acceleration), 0.05°/s (angular velocity) I2C Address: 0x68
Three-axis geomagnetic sensor (BMM150)	Accuracy: 0.3 μT Mounted on BMI270, magnetometer data obtained through BMI270
Camera	Sensor Chip: OV3660 Max Frame Rate: 30 fps Output Format: RAW RGB, RGB565/555/444, CCIR656, YCbCr422, and compression Aperture: F2.4 Resolution: 3MP (3 megapixels) Focal Length: 1.8±5%mm Field of View: FOV 120°
Infrared IR	Infrared emission distance at ∠180°: 12.46m (In the absence of obstruction)
Sleep Current	GPIO-5V Power: DC 5V@11.63 uA Grove-5V Power: DC 5V@10.75 uA USB-5V Power: DC 5V@92.50 uA (including PD resistor loss)
Reserved GPIO at Bottom	G5/G6/G7/G8/G38/G39
Operating Temperature	0 ~ 40°C
Product Size	26.4 x 24.0 x 22.5mm
Product Weight	10.8g
Package Size	48.0 x 46.4 x 28.5mm
Gross Weight	17.3g

Learn

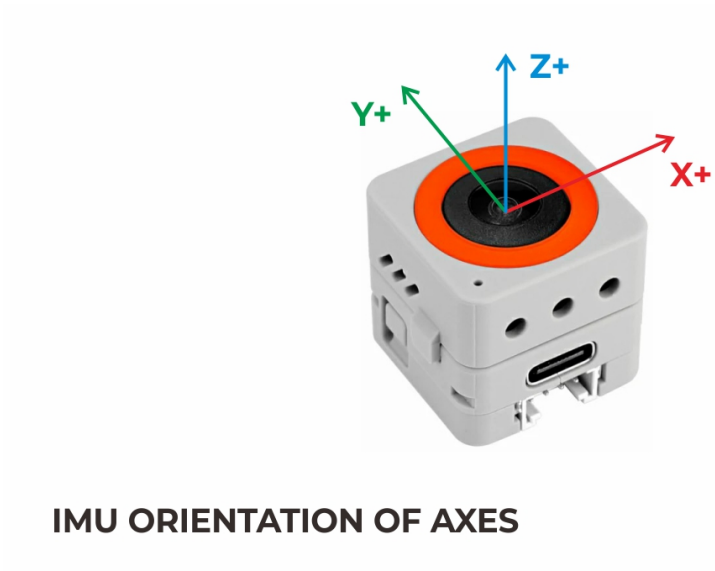
Download Mode

To flash firmware, press and hold the reset button (about 2 seconds) until the internal green LED lights up, then release the button.

After release, the green LED turns off, indicating that the device has entered download mode and is ready for flashing.

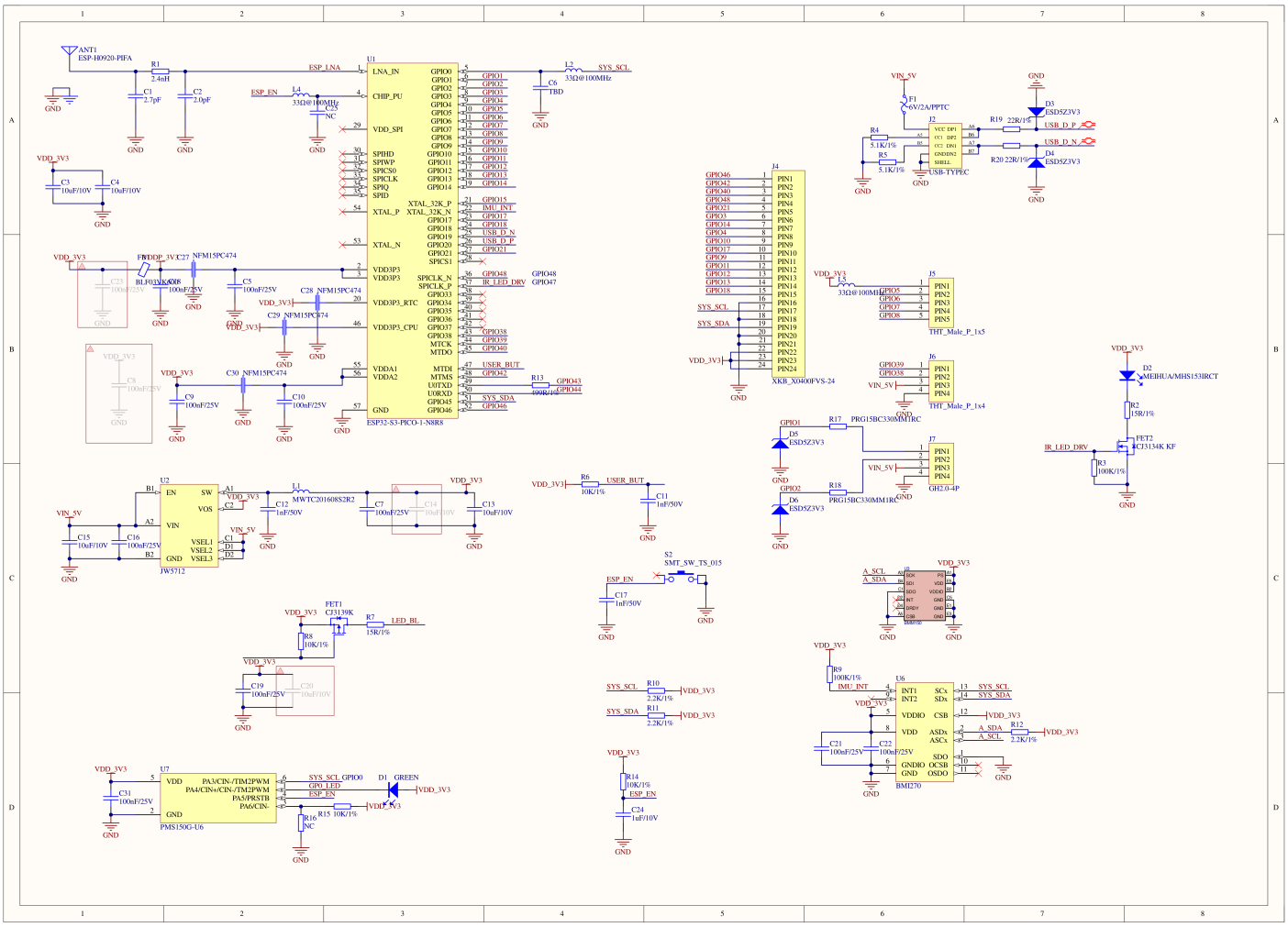


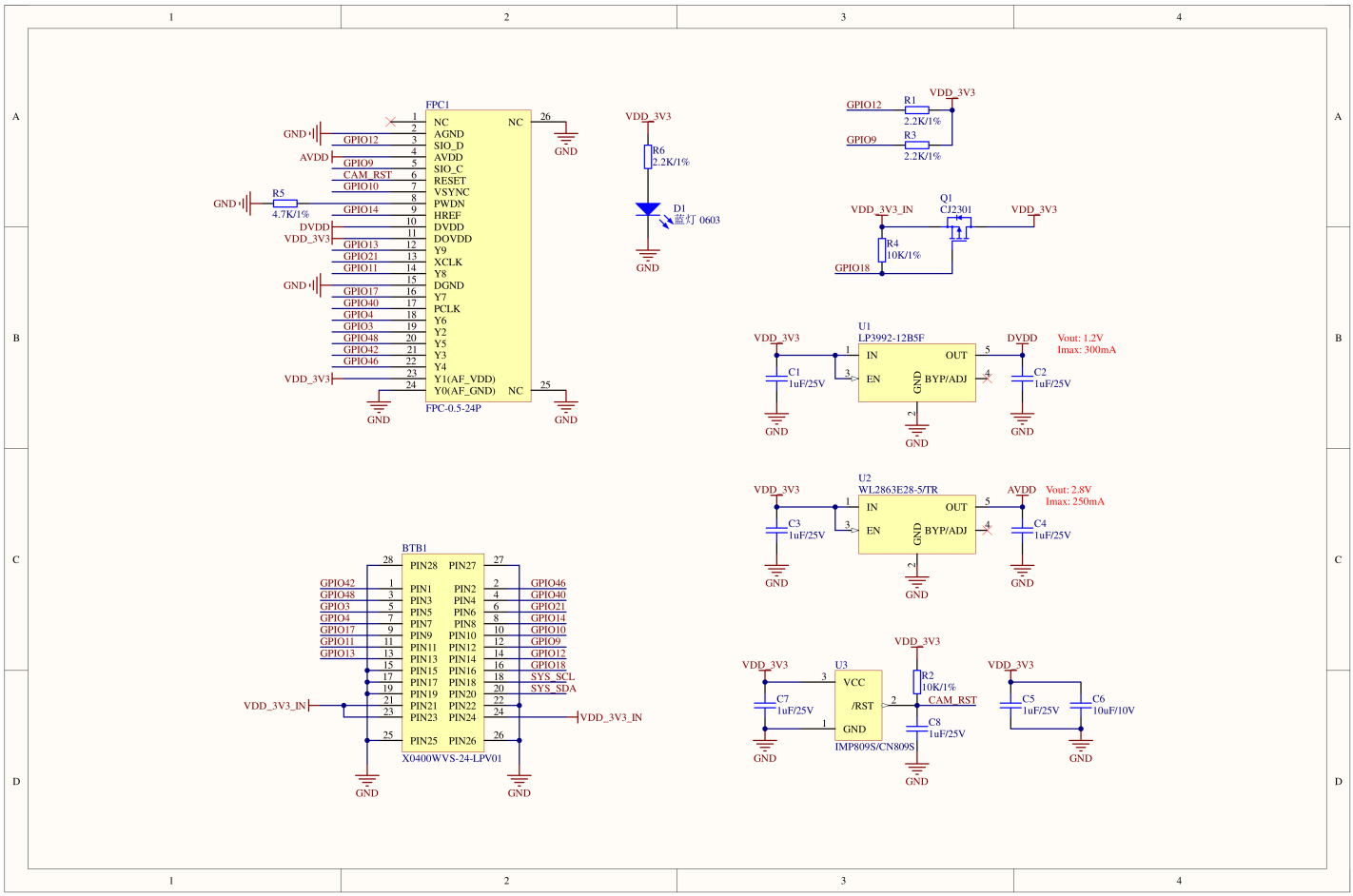
IMU Triaxial Direction Schematic Diagram



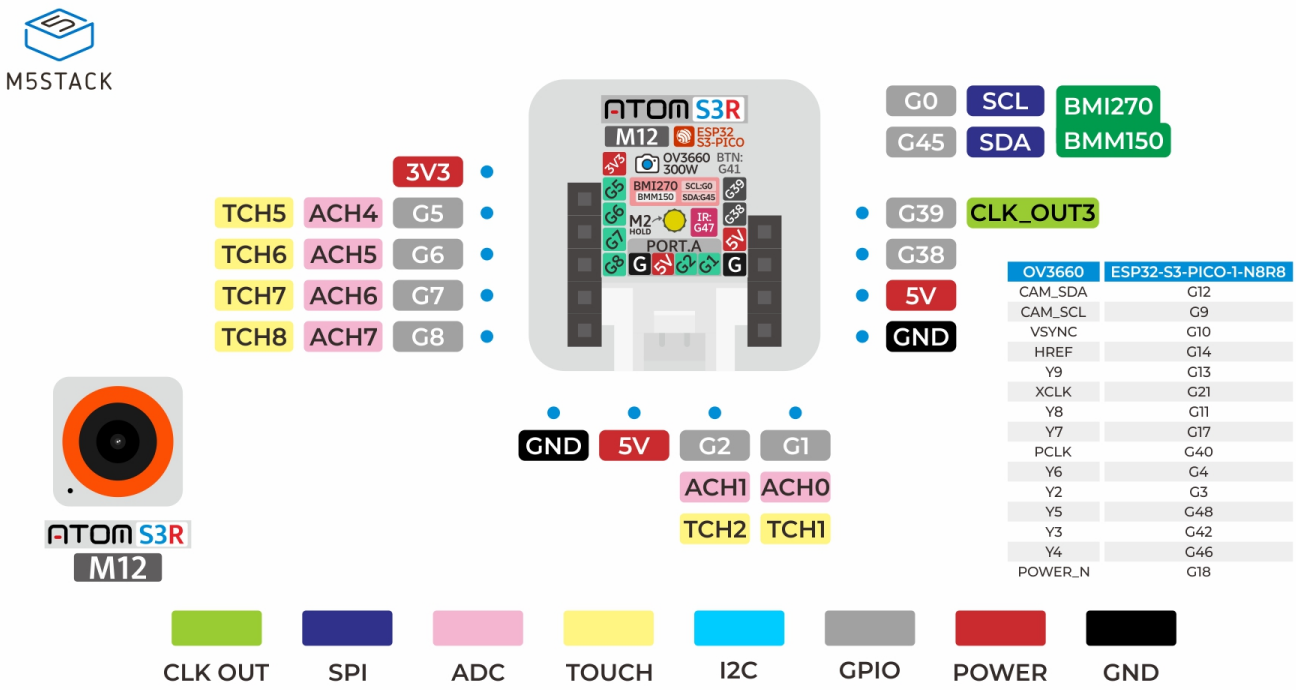
Schematics

- AtomS3R-M12 main board Schematics PDF
- AtomS3R-M12 ext board Schematics PDF


ANT1
ESP-H0920-PIFA
ESP32-S3-PICO-1-N8R8
JW5712
PMS150G-U6
FPC-0.5-24P
XKB_X0400FVS-24
THT_Male_P_1x5
THT_Male_P_1x4
GH2.0-4P
USB-TYPEC
MEIHUA/MHS153IRCT
SMT_SW_TS_015
MWTC201608S2R2
PRG15BC330MM1RC
ESD5Z3V3
IMU_INT
BMI270
LED_BL
IR_LED_DRV
USER_BUT
SYS_SCL
SYS_SDA
ESP_EN
ESP_LNA
VDD_3V3
VIN_5V
GND


FPC1
FPC-0.5-24P
D1
蓝灯 0603
U1
LP3992-12B5F
Vout: 1.2V
Imax: 300mA
U2
WL2863E28-5/TR
Vout: 2.8V
Imax: 250mA
Q1
CJ2301
U3
IMP809S/CN809S
BTB1
X0400WVS-24-LPV01
VDD_3V3
VDD_3V3_IN
GND

PinMap



BMI270 & IR

ESP32-S3-PICO-1-N8R8	G0	G45	G47
BMI270	SYS_SCL	SYS_SDA	
IR			IR_LED_DRV

BMM150

BMI270	BMI270_ASDx	BMI270_ASCx
BMM150	A_SDA	A_SCL

BMM150 Mounted on BMI270

Integrating BMM150 via BMI270’ s Sensor Hub auxiliary I2C interface for unified 9-axis sensor data collection

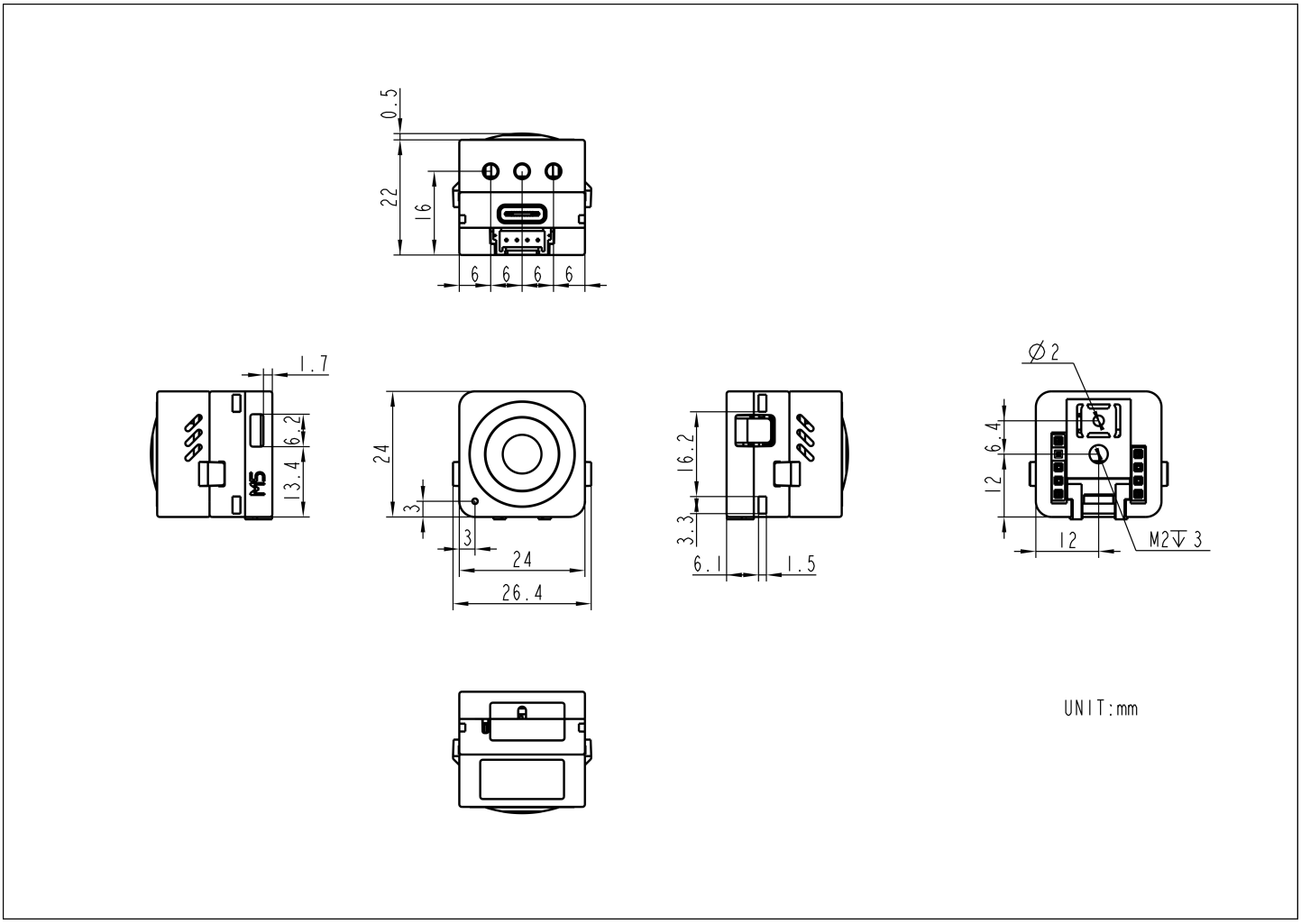
OV3360(M12)

OV3360(M12)	ESP32-S3-PICO-1-N8R8
CAM_SDA	G12
CAM_SCL	G9
VSYNC	G10
HREF	G14
Y9	G13
XCLK	G21
Y8	G11
Y7	G17
PCLK	G40
Y6	G4
Y2	G3
Y5	G48
Y3	G42
Y4	G46
POWER_N	G18

HY2.0-4P

HY2.0-4P	Black	Red	Yellow	White
PORT.CUSTOM	GND	5V	G2	G1

Model Size



Structure

- AtomS3R-M12 Structure Files

Datasheets

- ESP32-S3-PICO-1-N8R8
- OV3660
- BMI270
- BMM150

Softwares

Arduino

- AtomS3R-M12 Arduino Quick Start
- AtomS3R-M12 Web Camera Example

UiFlow2

- AtomS3R-M12 UiFlow2 Quick Start

Home Assistant

- Home Assistant Integration

PlatformIO

```
[env:m5stack-atoms3r]
platform = espressif32@6.7.0
board = esp32-s3-devkitc-1
framework = arduino
board_build.arduino.memory_type = qio_opi
build_flags =
    -DESP32S3
    -DBOARD_HAS_PSRAM
    -mfix-esp32-psram-cache-issue
    -DCORE_DEBUG_LEVEL=5
    -DARDUINO_USB_CDC_ON_BOOT=1
    -DARDUINO_USB_MODE=1
lib_deps =
    M5Unified=https://github.com/m5stack/M5Unified
```

ESP-IDF

- AtomS3R-M12 Factory Firmware

> **Factory Firmware Functionality**
>
> The factory firmware provides `UVC` functionality and `Wi-Fi image transmission`.
> UVC functionality allows direct USB connection to a computer, where you can preview images using a PC camera application.
> For Wi-Fi image transmission, connect to the device's AP: AtomS3R-M12-WiFi, then enter 192.168.4.1 in a browser to access the image transmission preview page.

Easyloader

Easyloader	Download Link	Note
AtomS3R-M12 User Demo Easyloader	download	/

Video

Product Comparison

Product Comparison		
	AtomS3R-M12	**AtomS3R-Cam**
Camera	**Image Sensor**: OV3660 **Max Frame Rate**: 30 fps **Output Format**: RAW RGB, RGB565/555/444, CCIR656, YCbCr422, and compression **Aperture**: F2.4 **Resolution**: 3MP (3 megapixels) **Focal Length**: 1.8 ± 5% mm **Field of View (FOV)**: 120°	**Image Sensor**: GC0308 **Max Frame Rate**: 30 fps **Output Format**: YCbCr4:2:2, RGB565, Raw Bayer **Aperture**: F2.6 **Resolution**: 0.3MP **Sensor Size** (Camera Diagonal Length): 1/6.5'' **Focal Length**: 2.43mm **Field of View (FOV)**: 57.6°
Resolution	3MP	0.3MP
Communication	Supports UVC and Wi-Fi communication methods	Supports UVC and Wi-Fi communication methods
SoC	ESP32-S3-PICO-1-N8R8	ESP32-S3-PICO-1-N8R8
Memory	8MB Flash + 8MB PSRAM	8MB Flash + 8MB PSRAM
Sensors	BMI270 + BMM150	BMI270 + BMM150
Antenna	Enhanced 3D antenna	Enhanced 3D antenna

To compare information on the Atom series products, you can visit the Product Selection Table, check the target products, and get the comparison results. The selection table covers key information such as core parameters and functional features, and supports comparison of multiple products simultaneously.